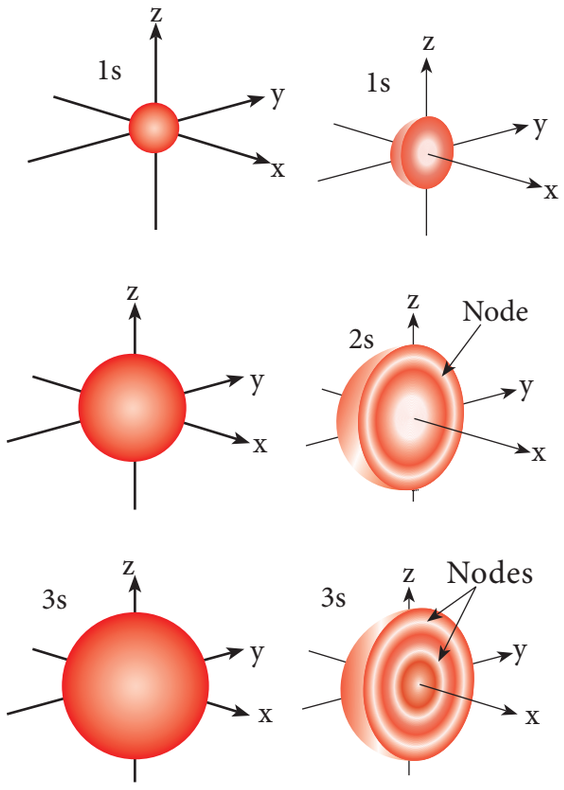
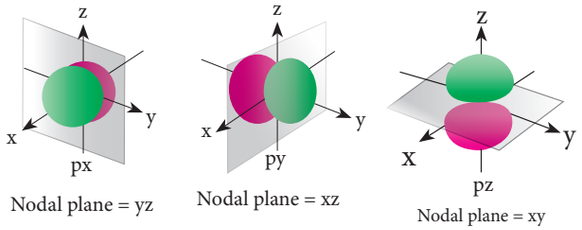



**Figure 2.7 Shapes of 1s, 2s and 3s orbitals**

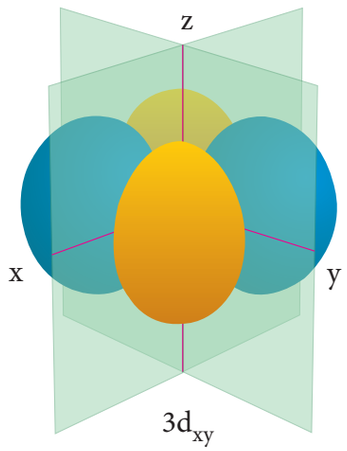
For p orbitals $l$=1 and the corresponding m values are -1, 0 and +1. The angular distribution functions are quite complex and are not discussed here. The shape of the p orbital is shown in Figure 2.8. The three different m values indicates that there are three different orientations possible for p orbitals. These orbitals are designated as $p_x$, $p_y$ and $p_z$ and the angular distribution for these orbitals shows that the lobes are along the x, y and z axis respectively. As seen in the Figure 2.8 the 2p orbitals have one nodal plane.

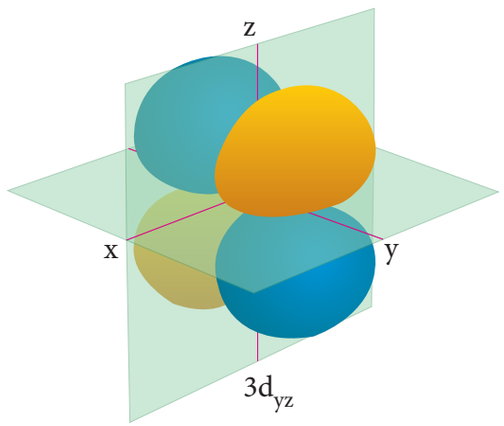



(b) Cartoon representations of 2p orbitals

**Figure 2.8 Shapes of 2p orbitals**

For 'd' orbital $l$ = 2 and the corresponding m values are -2, -1, 0 +1,+2. The shape of the d orbital looks like a 'clover leaf'. The five m values give rise to five d orbitals namely $d_{xy}$ , $d_{yz}$, $d_{zx}$, $d_{x^2-y^2}$ and $d_{z^2}$. The 3d orbitals contain two nodal planes as shown in Figure 2.9.






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